MAT1001 Midterm Examination

Saturday, October 26, 2024

Time: 9:00 - 11:00 AM

Notes and Instructions

1. No books, no notes, no dictionaries, no calculators, and no phones.

2. The maximum possible score of this examination is ***129****.*

3. There are ***13*** *problems (with parts) in total.*

*4. The symbol [****N****] at the beginning of a question indicates that the question is worth N points.*

5. Write down your solutions on the ***answer book****.*

6. Show your intermediate steps ***except Questions 1 and 2*** *— answers without intermediate steps will receive minimal (or even zero) marks.*

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MAT1001 Midterm Questions

1. [**10**] True or False? No need to show intermediate steps.

   (i) If f is continuous on (a, b) then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in (a, b). F

   (ii) If a car's speedometer reads 30 km/hour at 2:00 PM and it reads 50 km/hour at 2:10 PM, then at some moment between 2:00 and 2:10 PM the car's acceleration is exactly 120 km/(hour)^2. (Assume that the car's velocity is differentiable.) T

   (iii) $f(x) = x^3 + 2x + \tan x$ does not have any local maximum or minimum values. $3x^2+2 + \sec^2 x > 0$ T

   (iv) If $x = 1$ is a critical point of a function $f(x)$, $f' < 0$ on the interval $(0, 1)$, and $f' > 0$ on the interval $(1, 2)$, then $f(1)$ is a local minimum of f. $f'(1)$ $\frac{0}{0}$ = DNE F

   (v) Supposing the function $f(x)$ is continuous over the interval $[a, b]$ except at a point $c \in [a, b]$. Then, we can have

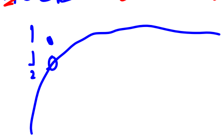
   $$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

   even though the discontinuity at point c is a finite jump. T

2. [**18**] Short questions. No need to show intermediate steps.

   (i) Is the following function continuous at $x = 1$? If not, state the type of discontinuity. NOPE / ~~HOLE~~ REMOVABLE

   $$f(x) = \begin{cases} \frac{x^2-x}{x^2-1}, & \text{if } x \neq 1, \\ 1, & \text{if } x = 1. \end{cases} \qquad \frac{x}{x+1}$$

   (ii) Determine the vertical asymptotes of $f(x) = \dfrac{x}{x^2 - 1}$. $x = 1, -1$

   (iii) Given a function of f, what is the definition of its derivative $f'(x)$ at the point x in its domain?

   slope of $y = f(x)$ at point x

   $\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

$y(u)$ $\qquad u(x)$

$\rightarrow \left.\frac{dy}{du}\right|_{u=1} = 10$, $u(26) = 1$ $\quad \left.\frac{du}{dx}\right|_{x=26} = 2024$

(iv) Let y be a function of u, and assume y is differentiable at $u = 1$ where the (instantaneous) rate of change of y with respect to u is 10; let us further assume that u in turn is a function of x such that $u = 1$ when $x = 26$, and u is differentiable at $x = 26$ where the (instantaneous) rate of change of u with respect to x is 2024. Find the (instantaneous) rate of change y with respect to x when $x = 26$. $\quad 20240$

$3x^2 - 1$

(v) We wish to find the solution to the equation $f(x) = x^3 - x - 5 = 0$ using Newton's method. Supposing we take the initial estimate $x_0 = 0$, then we have for the next estimate

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0 - \frac{-5}{-1}$

(A) $x_1 = -1$
(B) $x_1 = -5$
(C) $x_1 = -7$
(D) $x_1 = 4$

(vi) Choose the correct answer: If f is continuous and we know that $f(0) = -1$ and $f(2) = 1$, then

(A) There is a unique solution of the equation $f(x) = 0$. X
(B) $-1 < f(x) < 1$ for any $x \in (0, 2]$. X
(C) $f(1) = 0$. X
(D) There is at least one solution $x \in (-1, 2)$ of $f(x) = 0$.

3. [**6**] Find the limit $\lim_{x\to\infty} \left(\sqrt{x^2 + x} - \sqrt{x^2 - x}\right)$.

$= \lim_{x\to\infty} \frac{2x}{\sqrt{x^2+x} + \sqrt{x^2-x}} = \lim_{x\to\infty} \frac{2x}{x\sqrt{1+\frac{1}{x}} + x\sqrt{1-\frac{1}{x}}} = \frac{2x}{x+x} = 1$

4. [**12**] Let a be a nonzero real number and g be a function satisfying $|g(x)| \leq 2$ for all $x > 0$. Consider the function

$$f(x) = \begin{cases} \frac{\sin(a^2 x) - x}{x}, & \text{if } x < 0, \\ a + 1, & \text{if } x = 0, \\ \cos^2\left(x + \frac{\pi}{2}\right) g\left(\frac{1}{x}\right), & \text{if } x > 0. \end{cases}$$

$\frac{\sin a^2 x}{x} = a^2 -$

$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} \frac{\sin(a^2x) - x}{x}$
$= \lim_{x\to 0^-} \frac{\sin(a^2x)}{x} - \lim_{x\to 0} \frac{x}{x}$
$= a^2 \lim_{x\to 0^-} \frac{\sin(a^2x)}{a^2x} - 1$
$= a^2 \cdot 1 - 1 = a^2 - 1$

(i) Find $\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0^+} f(x)$.

$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \cos^2\left(x + \frac{\pi}{2}\right) g\left(\frac{1}{x}\right) = \lim_{x\to 0^+} (-\sin x)^2 g\left(\frac{1}{x}\right)$

(ii) Determine all possible values of a for which $\lim_{x\to 0} f(x)$ exists. What are the values of a, if any, such that f is continuous at $x = 0$?

$-2 \leq g\left(\frac{1}{x}\right) \leq 2$

$-2 \lim_{x\to 0^+} \sin^2 x \leq \lim_{x\to 0^+} \sin^2 x\, g\left(\frac{1}{x}\right) \leq 2 \lim_{x\to 0^+} \sin^2 x$

$0 \leq \lim_{x\to 0^+} \sin^2 x\, g\left(\frac{1}{x}\right) \leq 0$

$\rightarrow \lim_{x\to 0^+} \sin^2 x\, g\left(\frac{1}{x}\right) = 0$

$\lim_{x\to 0^-} f(x) = f(0) = \lim_{x\to 0^+} f(x)$

$a^2 - 1 = a + 1 = 0$

$a = -1 \rightarrow$ True

5. [**15**] Had Galileo dropped a cannonball from the Tower of Pisa, 98 m above the ground, the ball's height above the ground t seconds into the fall would have been $s(t) = 98 - 4.9t^2$ meters.

$\text{velocity} = -9.8t \ \text{m/s}$ $\quad$ $\text{speed} = 9.8t \ \text{m/s}$ $\quad$ $\text{acceleration} = -9.8 \ \text{m/s}^2$

(i) What would have been the ball's velocity, speed, and acceleration at time t? (Hint: please remember to write down the unit.)

(ii) About how long would it have taken the ball to hit the ground? $2\sqrt{5} \approx 4.47$ s

(iii) What would have been the ball's velocity at the moment of impact?

$V(\sqrt{20}) = -9.8 \cdot \sqrt{20}$

6. [**12**] Compute the derivative of each of the following functions.

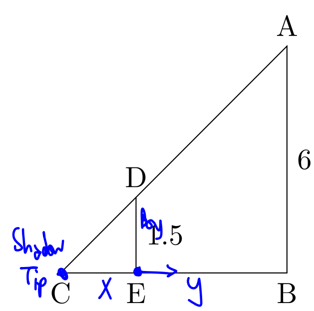
$0 + 2\cos 2X - 4X\sin(2X) + \dfrac{\sqrt{1+X^2}(3\sec^2 X) - 3\tan X\left(\frac{X}{\sqrt{1+X^2}}\right)}{1+X^2}$

(i) $5\pi^2 + 2x\cos(2x) + \frac{3\tan x}{\sqrt{1+x^2}}$.

(ii) $y^4 - 4y^2 = x^4 - 9x^2$, find $\frac{dy}{dx}$ at $(x, y) = (3, 2)$.

$4y^3 \frac{dy}{dX} - 8y \frac{dy}{dX} = 4X^3 - 18X$ $\qquad$ $\frac{dy}{dX} = \frac{4X^3 - 18X}{4y^3 - 8y} = \frac{4 \cdot 27 - 18 \cdot 3}{4 \cdot 8 - 8 \cdot 2} = \frac{54}{16} = \frac{27}{8}$

7. [**8**] A boy 1.5 m tall is walking towards a 6 m lamppost at the rate of 2 m per second, as shown in Figure 1, where B indicates the position of the lamppost, and E indicates the position of the boy. How fast is the tip of the boy's shadow (cast by the lamp) moving?



$\frac{dy}{dt} = -2$

$\frac{X}{1.5} = \frac{X+y}{6}$

$\frac{1}{1.5}\frac{dX}{dt} = \frac{1}{6}\left(\frac{dX}{dt} + \frac{dy}{dt}\right)$

$\frac{2}{3}\frac{dX}{dt} = \frac{1}{6}\frac{dX}{dt} - \frac{1}{3}$

$\frac{1}{3} = -\frac{1}{2}\frac{dX}{dt} \rightarrow \frac{dX}{dt} = -\frac{2}{3}$

$\frac{d(X+y)}{dt} = -\frac{8}{3}$ m/s

Figure 1: Boy at E walking towards a lamppost at B

8. [**6**] A coat of paint of thickness 0.01 cm is applied to the faces of a cube whose edge is 10 cm, thereby producing a slightly larger cube. Use a differential to estimate the number of cubic centimetres of paint used.

$V = (X + 2dX)^3 - X^3$

$= \cancel{X^3} + 6X^2 dX + \cancel{12X dX^2} + \cancel{8dX^3} - \cancel{X^3}$

$= 6X^2 dX \approx 6 \cdot 10^2 \cdot 0.01 = 6 \text{ cm}^3$

9. [**12**] Consider the function

$f''(x) = 30x^4 - 36x^2 \quad x = \cancel{\sqrt{\tfrac{6}{5}}}, -\sqrt{\tfrac{6}{5}}$ Inflection

$\cancel{\sqrt{\tfrac{6}{5}}} > 1$

$f'(x) = 6x^5 - 12x^3 \quad x = -\sqrt{2}, 0, 1$ Critical

$$f(x) = \begin{cases} x^6 - 3x^4, & x < 1 \\ -(\sqrt{x-1} + 2), & x \geq 1 \end{cases}$$

0 / DNE

(i) Find all critical points of f. $\quad f'(x) = -\dfrac{1}{2\sqrt{x-1}}$

(ii) Find all inflection points of f. $\quad f''(x) = \dfrac{1}{4\sqrt{(x-1)^3}} > 0 \quad \forall x \geq 1$

$A^2(0) = A^2(2) = 0$

10. [**8**] Find the largest area for a rectangle whose diagonal has length $\sqrt{2}$.

$A = a\sqrt{2-a^2} \rightarrow A^2 = 2a^2 - a^4 \qquad \therefore A^2(1) = 2(1^2) - 1^4 = 1 \rightarrow A(1) = 1$

$\dfrac{d(A^2)}{da} = 4a - 4a^3 = 0 \quad \therefore a = \cancel{-1}, \cancel{0}, 1$

11. [**6**] Solve the initial value problem $\begin{cases} \dfrac{dy}{dx} = 1 - \cos x, \\ y(0) = 1. \end{cases}$. $\rightarrow y = x - \sin x + C$

$1 = 0 - 0 + C \rightarrow C = 1$

$\therefore y = x - \sin x + 1$

12. [**8**] Using known methods for computing areas, find the mean value (average value) of the function $dx = a\cos\theta\, d\theta \quad x = a\sin\theta$

$$f(x) = \sqrt{a^2 - x^2}$$

$\displaystyle\int_{-a/2}^{a/2} f(x)\,dx = \int_{-\pi/6}^{\pi/6} a^2\cos^2\theta\, d\theta$

$\displaystyle = a^2 \int_{-\pi/6}^{\pi/6} \frac{\cos 2\theta + 1}{2}\, d\theta$

$\displaystyle = a^2 \left(\frac{\sin 2\theta}{4} + \frac{\theta}{2} \right)\Bigg|_{-\pi/6}^{\pi/6}$

$\displaystyle = a^2 \left(\frac{\sqrt{3}}{4} + \frac{\pi}{6} \right)$

over the interval $-\frac{a}{2} \leq x \leq \frac{a}{2}$ (where $a > 0$).

13. [**8**] A particle travels along the x-axis with the velocity

$$v(t) = t(1-t) \text{ m/s}, \quad t \geq 0.$$

$v(t) = t - t^2$

Show that over the interval $0 \leq t \leq 1$, the distance it covers is not more than 0.25 m.

$0 = v'(t) = 1 - 2t \rightarrow t = 0.5$ max pt.

$v''(t) = -2$

$\therefore v(0.5) = 0.5 - 0.5^2 = 0.25$

$\therefore \displaystyle\int_0^1 v(t)\,dt \leq 0.25(1-0) = 0.25$